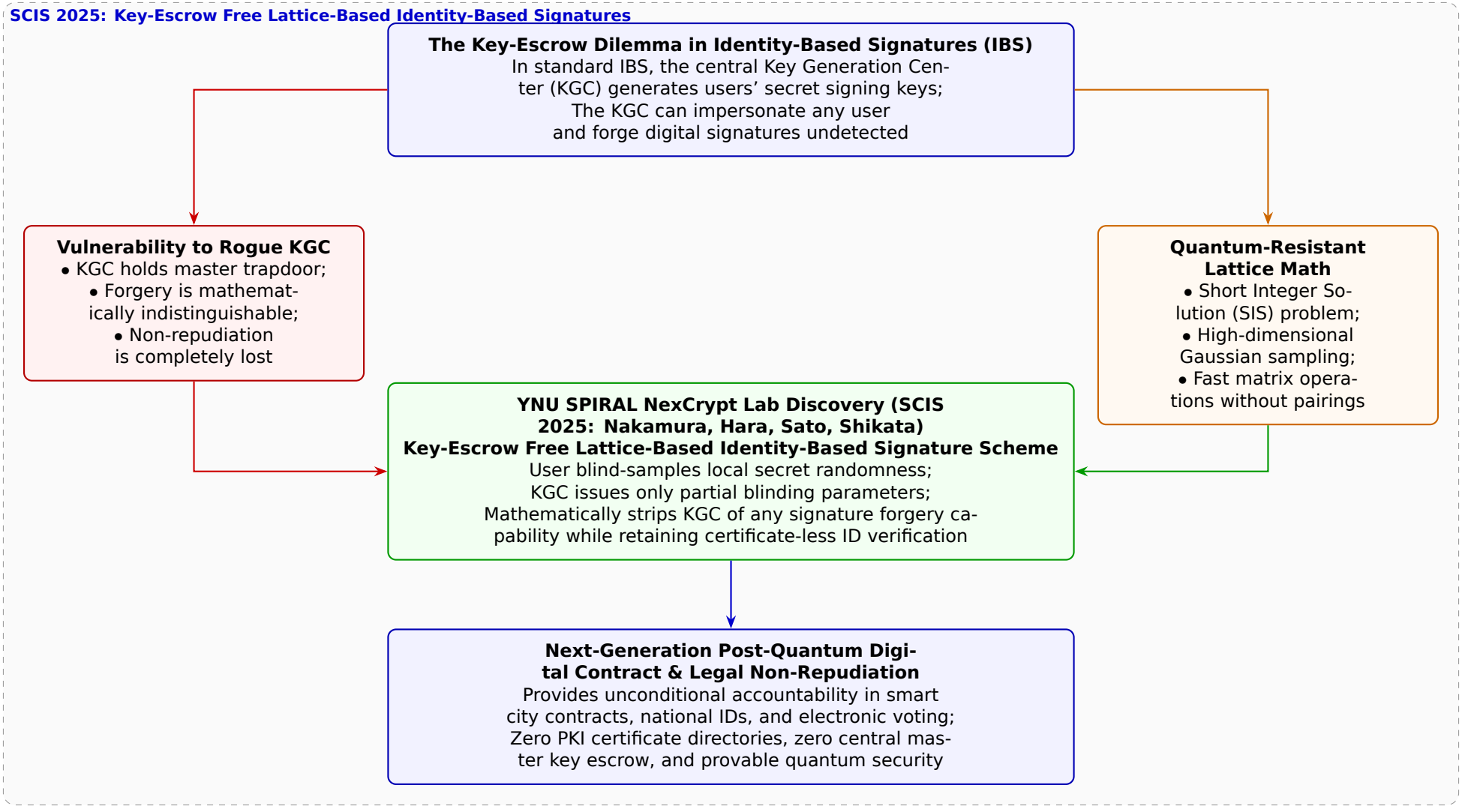
SCIS 2025: Key-Escrow Free Lattice-Based Identity-Based Signatures
The Key-Escrow Dilemma in Identity-Based Signatures (IBS)
In standard IBS, the central Key Generation Center (KGC) generates users' secret signing keys;
The KGC can impersonate any user
and forge digital signatures undetected
Vulnerability to Rogue KGC
• KGC holds master trapdoor;
• Forgery is mathematically indistinguishable;
• Non-repudiation
is completely lost
Quantum-Resistant Lattice Math
• Short Integer Solution (SIS) problem;
• High-dimensional Gaussian sampling;
• Fast matrix operations without pairings
YNU SPIRAL NexCrypt Lab Discovery (SCIS 2025: Nakamura, Hara, Sato, Shikata)
Key-Escrow Free Lattice-Based Identity-Based Signature Scheme
User blind-samples local secret randomness;
KGC issues only partial blinding parameters;
Mathematically strips KGC of any signature forgery capability while retaining certificate-less ID verification
Next-Generation Post-Quantum Digital Contract & Legal Non-Repudiation
Provides unconditional accountability in smart city contracts, national IDs, and electronic voting;
Zero PKI certificate directories, zero central master key escrow, and provable quantum security